Germain Vivier-Ardisson

PhD Student in Machine Learning and Operations Research

Born: 16/05/2001 | gvivier@google.com
Languages: French (native), English (fluent) | germain.vivier-ardisson@enpc.fr
Last edited on: August 27, 2026 | https://germainvivierardisson.github.io/

Positions and Education

Co-Advised PhD, *Google DeepMind & CERMICS, ENPC* — Paris
Machine learning & operations research. Advised by A. Parmentier and M. Blondel. — Jan 2025 - (Current)

Research Internship, *CERMICS, ENPC* — Marne-la-Vallée
Combinatorial optimization heuristics as layers. Advised by A. Parmentier and M. Blondel. — Apr 2024 - Aug 2024

Master of Science (M2), *M2A (Learning and Algorithms), Sorbonne Université* — Paris
Learning theory, convex and non-convex optimization, generative modeling, GPU programming. — 2023 - 2024

Research Internship, *MAGI, Polytechnique Montréal* — Montréal, Canada
Counterfactual explanations for structured prediction. Advised by T. Vidal. — Mar 2023 - Aug 2023

Research & Development Internship, *Quantmetry (now part of Capgemini)* — Paris
Benchmark and implementation of hybrid machine learning / operations research algorithms. — Mar 2023 - Aug 2023

Master of Science/Engineering (cycle ingénieur polytechnicien), *Ecole polytechnique* — Palaiseau
Applied mathematics, computer science, physics. — 2020 - 2024

Bachelor of Science (classes préparatoires PTSI/PT), *Lycée Privé Sainte-Geneviève* — Versailles
Mathematics, physics, engineering. Ranked 4th at Ecole polytechnique's entrance exam. — 2018 - 2020

Teaching Experience

- **Teaching Assistant**, Introduction to convex optimization and linear programming at ENPC (1A) — Fall 2024
- **Oral Examiner in Mathematics** at Lycée Saint-Louis (Colleur en filière PSI) — Sep 2023 - Jun 2024

Publications

[1] **G. Vivier-Ardisson**, L. Demonet, A. Parmentier, and M. Blondel, "Regularized Large Neighborhood Search", *arXiv preprint arXiv:2606.02294*, 2026.

[2] **G. Vivier-Ardisson**, M. E. Sander, A. Parmentier, and M. Blondel, "Differentiable Knapsack and Top-k Operators via Dynamic Programming", *arXiv preprint arXiv:2601.21775*, 2026.

[3] M. Blondel, M. E. Sander, **G. Vivier-Ardisson**, T. Liu, and V. Roulet, "Autoregressive Language Models are Secretly Energy-Based Models: Insights into the Lookahead Capabilities of Next-Token Prediction", *International Conference on Machine Learning*, 2026.

[4] **G. Vivier-Ardisson**, M. Blondel, and A. Parmentier, "Learning with Local Search MCMC Layers", *Transactions on Machine Learning Reasearch (Featured & J2C Certifications)*, 2026.

[5] **G. Vivier-Ardisson**, A. Forel, A. Parmentier, and T. Vidal, "CF-OPT: Counterfactual Explanations for Structured Prediction", *International Conference on Machine Learning*, 2024.